Complex variable

MTH 632

EX#1

Q#01 verify

i) $(\sqrt{2} - i) - i(1 - \sqrt{2}i) = -2i$

$= \sqrt{2} - i - i + \sqrt{2}i^2$

$\therefore i^2 = -1$

$= \sqrt{2} - 2i - \sqrt{2}$

$= -2i$

$= L.H.S$

ii) $(2, -3)(-2, 1) = (-1, 8)$

Solution:

$= (2, -3)(-2, 1)$

$= (-4 + 3, +6 + 2)$

$= (-1, 8)$

iii) $(3,1)(3,-1)\left(\frac{1}{5}, \frac{1}{10}\right) = (2,1)$

$= (3,1)(3,-1)$

$= (9+1, 3-3)$

$= (10, 0)\left(\frac{1}{5}, \frac{1}{10}\right)$

$= \left(\frac{10}{5} - 0, 0 + \frac{10}{10}\right)$

Since $(x_1, y_1)(x_2, y_2)$

$= (x_1 x_2 - y_1 y_2, y_1 x_2 + x_1 y_2)$

Q#02

a) $Re(iz) = -Im\,z$

$Re[i(x+iy)] = Re(-y+ix)$

$= -y = -Im\,z$

b) $Im(iz) = Re\,z$

$Im(iz) = Im[i(x+iy)]$

$= Im(-y+ix)$

$= x$

$= Re\,z$

Q#03 Show That

$(1+z)^2 = 1+2z+z^2$

$(1+z)^2 = (1+z)(1+z)$

$= 1\cdot(1+z) + z\cdot(1+z)$

$= 1+z+z+z^2$

$= 1+2z+z^2$

Q#04 verify That each of The two numbers $z = 1 \pm i$ satisfies the equation $z^2 - 2z + 2 = 0$

if $z = 1 \pm i$ then

$$z^2 - 2z + 2 = (1 \pm i)^2 - 2(1 \pm i) + 2$$
$$= 1 + i^2 \mp 2i - 2 \mp 2i + 2$$
$$= 1 - 1 \mp 2i - 2 \mp 2i + 2$$
$$= \mp 2i - 2 \mp 2i + 2$$
$$= 0$$

Q#05 Prove that multiplication of complex numbers is commutative (as stated at the beginning of set)

To prove That multiplication is commutative, write.

$$z_1 z_2 = (x_1, y_1)(x_2, y_2)$$

$$= (x_1 x_2 - y_1 y_2, y_1 x_2 + x_1 y_2)$$
$$= (x_2 x_1 - y_2 y_1, y_2 x_1 + x_2 y_1)$$
$$= z_2 z_1$$

Q#06 verify

a) the associative law for addition of Complex numbers.

$$(z_1+z_2)+z_3 = [(x_1,y_1)+(x_2,y_2)]+(x_3,y_3)$$

$$= (x_1+x_2, y_1+y_2)+(x_3,y_3)$$

$$= (x_1+x_2)+x_3, (y_1+y_2)+y_3$$
$$= x_1+(x_2+x_3), y_1+(y_2+y_3)$$
$$= (x_1,y_1)+(x_2+x_3, y_2+y_3)$$
$$= (x_1,y_1)+((x_2,y_2)+(x_3,y_3))$$
$$= z_1+(z_2+z_3)$$

b) the distributive law.

the distributive law write

$$z(z_1+z_2) = (x,y)((x_1,y_1)+(x_2,y_2))$$
$$= (x,y)(x_1+x_2, y_1+y_2)$$
$$= xx_1+xx_2-yy_1-yy_2, yx_1+yx_2 + xy_1+xy_2)$$
$$= xx_1-yy_1+xx_2-yy_2, yx_1+xy_1 + yx_2+xy_2)$$
$$= (xx_1-yy_1, yx_1+xy_1)+(xx_2-yy_2, yx_2+xy_2)$$

$= (x,y)(x_1,y_1) + (x,y)(x_2,y_2)$

$= zz_1 + zz_2$

Q#09 Use $-1 = (-1,0)$ and $z = (x,y)$ to show that $(-1)z = -z$

$= (-1)z$

$= (-1,0)(x,y)$

$= (-x-0,\ 0+(-y))$

$= (-x,-y)$

$= -(x,y)$

$= -z$

Ittihad